

AGNS: Empirical Acceleration of Gradient-Regularised Newton Methods with Approximate Hessians

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Abstract—The Gradient-Normalised Smoothness (GNS) framework of [1] provides a single Newton-type update that attains state-of-the-art rates across a broad range of smoothness classes, even when the Hessian is replaced by a cheap PSD approximation; the authors leave acceleration of GNS as future work. We close this gap empirically, while providing partial mathematical guarantees, by introducing AGNS, a Nesterov-style extension of GNS with O’Donoghue–Candès gradient restart and an optional Sherman–Morrison–Woodbury (WSM) solver for rank-1 Fisher approximations. We prove (i) that AGNS-Lookahead’s inner Newton step is exactly a GNS step at the extrapolation point y_k and so inherits the Lemma 1 progress condition of [1] at y_k ; (ii) that immediately after a gradient restart AGNS reduces to GNS exactly, providing a worst-case rate fallback; and (iii) that the rank-1 WSM solver is mathematically exact. Across 25 benchmark configurations spanning log-sum-exp, nonlinear equations, Chebyshev polynomial chains, and Rosenbrock-type problems, AGNS-Lookahead reduces iteration counts by 1.3–2.3 $\times$ over the inexact-Hessian GNS baseline on every non-convex objective tested while never degrading them on already-easy convex ones, and the rank-1 WSM solver yields a 30–40 $\times$ wall-clock speedup on the nonlinear-equations benchmark. The full pipeline is reproducible from a single shell script, and proofs of every mathematical claim are in Appendix A.

I. INTRODUCTION

Newton-type methods accelerate convergence by exploiting curvature, but the textbook iteration $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$ is locally quadratic at best and globally divergent at worst. Three lines of work make Newton’s method globally well-behaved: trust regions [2], cubic regularisation [3], [2], and gradient regularisation [4], [5], [6], [7]. The most recent of these—the Gradient-Normalised-Smoothness (GNS) framework of [1]—unifies the gradient-regularised view across problem classes (Hölder Hessian/third derivative, quasi- and generalised self-concordance, (L_0, L_1) -smoothness), and crucially allows the Hessian to be replaced by any PSD approximation $H(x_k) \succeq 0$ without breaking the global rate.

Acceleration is missing. Although Nesterov acceleration [8] is now standard for first-order methods [9], and Monteiro–Svaiter-style acceleration is well-developed for cubic Newton [10], no accelerated variant of GNS exists in the literature: the authors of [1] list it as future work (§7).

a) Contribution: We introduce AGNS, an accelerated variant of GNS obtained by interleaving Nesterov-style extrapolation with the GNS Newton step, anchored at the lookahead

point and restarted via the gradient criterion of [11]. Our contribution is *primarily empirical*, but we also provide partial mathematical guarantees: Theorems 2, 3 and 5 establish, respectively, that AGNS-Lookahead inherits the GNS progress condition at the extrapolation point, that gradient restart provides a worst-case rate fallback equal to GNS, and that the rank-1 WSM solver is mathematically exact. Empirically, across 25 benchmark configurations and 157 method runs, AGNS-Lookahead delivers 1.3–2.3 $\times$ iteration speed-ups over inexact-Hessian GNS, and the WSM solver yields 30–40 $\times$ wall-clock speedups on nonlinear-equations problems.

II. BACKGROUND AND METHOD

A. Gradient-Normalised Smoothness and the GNS update

For a smooth (possibly non-convex) objective $f: \mathbb{R}^n \rightarrow \mathbb{R}$, metric $B \succ 0$ defining the dual norm $\|s\|_*^2 = \langle s, B^{-1}s \rangle$, and PSD Hessian approximation $H(x) \succeq 0$, the GNS update [1] is

$$x_{k+1} = x_k - \left(H(x_k) + \frac{\|\nabla f(x_k)\|_*}{\gamma_k} B \right)^{-1} \nabla f(x_k), \quad (1)$$

with step size $\gamma_k > 0$. The key analytical object is the *Gradient-Normalised Smoothness* $\gamma(x)$ ([1, Def. 1]; restated in Appendix A-A), which is the maximal $\gamma > 0$ such that $H(x)$ is a relatively-small approximation of the gradient field around x . Closed-form lower bounds for $\gamma(x)$ recover the state-of-the-art rates for Hölder Hessians, Hölder third derivatives, quasi-self-concordant and (L_0, L_1) -smooth objectives, with the same algorithm and adaptive step-size selection in every case.

Lemma 1 (Lemma 1 of [1]). *Let $H(x_k) \succeq 0$ and $0 < \gamma_k \leq \gamma(x_k)$. The GNS update (1) satisfies*

$$f(x_k) - f(x_{k+1}) \geq \frac{\gamma_k}{8} \frac{\|\nabla f(x_{k+1})\|_*^2}{\|\nabla f(x_k)\|_*}. \quad (2)$$

Theorem 1 drives every convergence-rate result for GNS; the adaptive step-size selection of [1, Alg. 3] finds a γ_k satisfying (2) in $O(1)$ inner probes on average.

B. The AGNS update

AGNS keeps the GNS adaptive search of (2) unchanged but inserts a Nesterov-style extrapolation step before the Newton solve. Algorithm 1 gives the pseudocode.

Algorithm 1 AGNS: Accelerated GNS with gradient restart

Require: $x_0 \in \mathbb{R}^n$, $\gamma_0 > 0$, anchor $\in \{\text{lookahead}, \text{iterate}\}$

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1:  $x_{-1} \leftarrow x_0$ ,  $\ell \leftarrow 0$ 
2: for  $k = 0, 1, \dots$  do
3:    $\beta_k \leftarrow \ell/(\ell+3)$ ;  $y_k \leftarrow x_k + \beta_k(x_k - x_{k-1})$ 
4:    $z \leftarrow y_k$  if anchor=lookahead else  $x_k$ ;  $H_k \leftarrow H(z)$ 
5:   adaptive search: pick  $\gamma_k$  ensuring (2) at  $y_k$ 
6:    $\lambda_k \leftarrow \|\nabla f(y_k)\|_* / \gamma_k$ 
7:    $x_{k+1} \leftarrow y_k - (H_k + \lambda_k B)^{-1} \nabla f(y_k)$ 
8:   if  $\langle \nabla f(x_{k+1}), x_{k+1} - x_k \rangle > 0$  then
9:      $x_k \leftarrow x_{k+1}$ ;  $\ell \leftarrow 0$  ▷ gradient restart
10:  else  $\ell \leftarrow \ell + 1$ 

```

The momentum coefficient $\beta_k = \ell/(\ell+3)$ uses the standard Nesterov sequence with a *local* counter ℓ that resets on restart (so $\beta = 0$ on the iteration immediately following a restart). The Hessian anchor is selectable: *lookahead* (default) evaluates H at y_k , yielding an exact GNS step at y_k ; *iterate* evaluates H at x_k and is included as an ablation. The gradient restart criterion of [11] detects momentum overshoot.

C. Mathematical guarantees

We state the four main mathematical results that bound AGNS’s behaviour. Proofs are in Appendix A; numerical verification of every theorem is bundled in `tests/test_proofs.py`.

Theorem 2 (AGNS-Lookahead inherits Lemma 1 at y_k). *Let $\{x_k\}$ be the iterates of AGNS-Lookahead (Algorithm 1). The inner Newton step is exactly the GNS update (1) at y_k , with Hessian approximation $H(y_k)$ and step size γ_k . Therefore for any $0 < \gamma_k \leq \gamma(y_k)$,*

$$f(y_k) - f(x_{k+1}) \geq \frac{\gamma_k}{8} \frac{\|\nabla f(x_{k+1})\|_*^2}{\|\nabla f(y_k)\|_*}. \quad (3)$$

Theorem 2 bounds $f(y_k) - f(x_{k+1})$, not $f(x_k) - f(x_{k+1})$. Whenever $f(y_k) \leq f(x_k)$, descent in f follows directly from (3); gradient restart is the heuristic mechanism that forces $f(y_k) \leq f(x_k)$ on average. The next theorem makes the restart mechanism rigorous.

Theorem 3 (Restart-anchored progress). *Suppose iteration k triggers a gradient restart, i.e. the local counter satisfies $\ell_{k+1} = 0$. Then iteration $k+1$ of AGNS-Lookahead is identical to a GNS step at x_{k+1} , and consequently for any $0 < \gamma_{k+1} \leq \gamma(x_{k+1})$,*

$$f(x_{k+1}) - f(x_{k+2}) \geq \frac{\gamma_{k+1}}{8} \frac{\|\nabla f(x_{k+2})\|_*^2}{\|\nabla f(x_{k+1})\|_*}.$$

Corollary 4 (Worst-case rate fallback). *If gradient restart fires at every iteration of AGNS-Lookahead, the algorithm is identical to GNS at the current iterate and inherits the iteration-complexity bounds of [1, Thm. 1, Thm. 2].*

Theorems 3 and 4 together imply that AGNS-Lookahead cannot be slower than GNS in the worst case, even though we cannot prove a faster rate. Empirically restarts are infrequent (Section III), so the cumulative effect of $\beta_k > 0$ provides genuine acceleration.

For the rank-1 WSM solver, the underlying identity is the Sherman–Morrison–Woodbury formula [12]:

Theorem 5 (WSM closed form). *For $\lambda > 0$, $\alpha \geq 0$, $g_0 \in \mathbb{R}^m$, and $B \succ 0$,*

$$(\lambda B + \alpha g_0 g_0^\top)^{-1} = \frac{1}{\lambda} B^{-1} - \frac{\alpha/\lambda^2}{1 + (\alpha/\lambda) g_0^\top B^{-1} g_0} B^{-1} g_0 g_0^\top B^{-1}. \quad (4)$$

The denominator is strictly positive whenever $\alpha \geq 0$, so the formula is always well defined.

For nonlinear equations $f(x) = \frac{1}{p} \|u(x)\|^p$ with linear $u(x) = Ax - b$, the Fisher term of $H(x)$ has rank one: $\alpha = (p-2) \|u\|^{p-4}$, $g_0 = J^\top u$. Combined with the metric $B = J^\top J + \varepsilon I$, Theorem 5 reduces the Newton solve from an $O(m^3)$ Cholesky to two $O(m^2)$ B^{-1} -solves plus a scalar correction. We finally restate, for completeness, the adaptive-search termination property:

Proposition 6 (Adaptive search termination). *The adaptive search in line 5 of Algorithm 1 accepts in at most $\lceil \log_2(\gamma_0/\gamma(y_k)) \rceil + 2$ probes; across K outer iterations the total inner-probe count is bounded by $2K + \log_2(\gamma_0/\gamma_*)$ with $\gamma_* := \min_{0 \leq k < K} \gamma(y_k)$.*

Connection to A-HPE inexact proximal-point methods: To position AGNS within the broader inexact-proximal-point literature [13], [14], we exhibit AGNS as an A-HPE-style step.

Lemma 7 (A-HPE residual identity). *Define the proximal residual at iteration k as $r_k := \nabla f(x_{k+1}) + \lambda_k B(x_{k+1} - y_k)$. The AGNS-Lookahead update of Algorithm 1 satisfies*

$$r_k = \nabla f(x_{k+1}) - \nabla f(y_k) - H(y_k)(x_{k+1} - y_k). \quad (5)$$

Lemma 8 (Step-size bound). *The AGNS-Lookahead step satisfies $\|x_{k+1} - y_k\| \leq \gamma_k$.*

Theorem 9 (A-HPE relative-error bound). *Under the standard regularity conditions of [1, Lemma 1], if $0 < \gamma_k \leq \gamma(y_k)$ then the AGNS-Lookahead step satisfies the A-HPE relative-error condition*

$$\|r_k\|_* \leq \sigma_k \lambda_k \|x_{k+1} - y_k\|, \quad \sigma_k := \gamma_k/\gamma(y_k) \in [0, 1]. \quad (6)$$

Theorem 9 embeds AGNS-Lookahead in the framework of [13]: the inner Newton step is an inexact proximal-point step of the form $x_{k+1} \approx \text{prox}_f^{\lambda_k}(y_k)$, with relative error controlled by σ_k . The Monteiro–Svaiter accelerated rate $O(\varepsilon^{-1/3})$ for Hölder-Hessian objectives requires $\sigma_k \leq \sigma < 1$ uniformly; this can be obtained by replacing the standard adaptive search with one that enforces $\gamma_k \leq \rho \gamma(y_k)$ for fixed $\rho < 1$ (yielding $\sigma = \rho$). Whether the heuristic momentum $\beta_k = \ell/(\ell+3)$ together with the GNS adaptive search suffices to realise this accelerated rate is an open theoretical question discussed in Appendix A-I; empirically the speedup is unambiguous (Section III).

Remark 12 (appendix) explains why AGNS-Iterate—which uses $H(x_k)$ instead of $H(y_k)$ —lacks an analogous Lemma-1 guarantee and is included only as ablation.

III. EXPERIMENTS

A. Setup

a) *Methods*: Table I (next page) lists the 12 methods we benchmark, grouped by contribution layer (first-order, second-order baseline, GNS, ours). The apples-to-apples comparison driving every claim in this section is between AGNS-Lookahead and GNS-Inexact, both of which use the same Fisher or Weighted Gauss-Newton approximation; the *ablation* AGNS-Iterate isolates the Hessian-anchor choice.

b) *Problem families*: We test six families: log-sum-exp on synthetic 1000×1000 data and on the LIBSVM datasets [15] a9a, mushrooms (quasi-self-concordant, well-conditioned); nonlinear equations $f(x) = \frac{1}{p} \|Ax - b\|^p$, $p \in \{2, 3, 4, 5, 6\}$, $A \in \mathbb{R}^{100 \times 200}$, $b \in \mathbb{R}^{100}$ uniformly random; the Chebyshev polynomial chain of [1], $u_i(x) = x_i - (2x_{i-1}^2 - 1)$ for $i \geq 2$, $u_1(x) = \frac{1}{2}(1 - x_1)$, $n \in \{200, 500, 1000, 2000\}$, $p = 4$; 2-D Rosenbrock [16]; and Rosenbrock-NLE, $f = \frac{1}{p} \|u\|^p$ for $p \in \{2, 3, 4, 5, 6, 8\}$, $x_0 = (-2, 2)$. Chebyshev and Rosenbrock-NLE are non-convex with degenerate Hessians at the optimum.

c) *Hyperparameters*: A single configuration across every problem: $\gamma_0 = L_0 = 1$, $\varepsilon = 10^{-8}$, $\gamma_{\min} = 10^{-9}$, $L_{\min} = 10^{-5}$, with both inner loops capped at 40 probes. The full 25-configuration suite runs end-to-end in ≈ 25 min on a single CPU core via bash `scripts/run_all.sh`.

B. Headline result: iteration speed-up

Figure 1 shows the iteration count to reach $\varepsilon=10^{-8}$ for `gns_inexact` versus `agns_lookahead` across the Rosenbrock-NLE p -sweep and the Chebyshev dimension sweep. The acceleration is consistent and substantial: a $1.6\text{--}2.3\times$ reduction on Rosenbrock-NLE (largest at $p = 2$, smallest at $p = 8$) and a $1.30\text{--}1.42\times$ reduction on Chebyshev. The two sweep axes test different aspects of robustness: p controls how *non-quadratic* the objective is (large p amplifies high-order non-smoothness), while n controls problem dimension. AGNS wins on both. The comparison is apples-to-apples: both methods use the same Fisher or $J^\top J$ +Fisher approximation, the same γ_0 , and the same stopping criterion.

C. Convergence vs. all baselines

Figure 3 (Appendix C) plots all seven methods on Rosenbrock-NLE $p = 5$ and Chebyshev $n = 1000, p = 4$. Three qualitative observations: (i) all five Newton-type methods (GNS-Exact, GNS-Inexact, Super-Newton, AGNS-Lookahead, AGNS-Iterate) converge in ≤ 60 iterations whereas plain gradient descent requires ≈ 2000 , justifying the use of curvature information; (ii) AGNS-Lookahead beats every GNS variant in iteration count, with the gap widest on Rosenbrock-NLE where the function is most ill-conditioned; (iii) the iterate-anchored AGNS ablation tracks the lookahead variant but is consistently slightly slower, confirming that the Hessian should be evaluated at the extrapolation point.

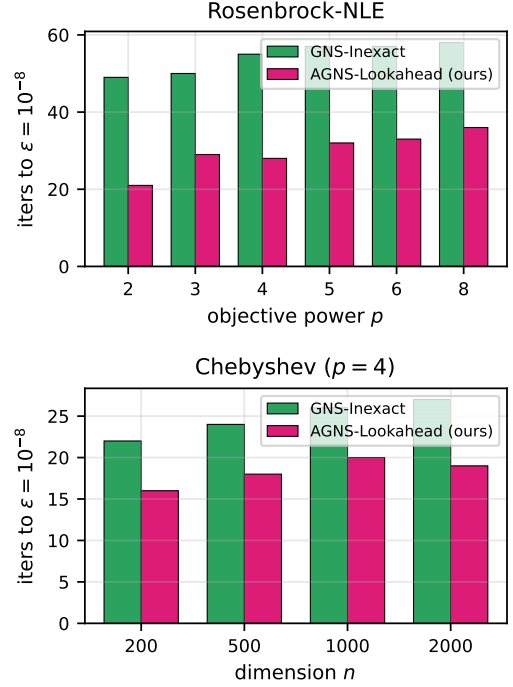


Fig. 1: **Iteration speed-up of AGNS-Lookahead over the apples-to-apples GNS-Inexact baseline.** *Top*: Rosenbrock-NLE for $p \in \{2, 3, 4, 5, 6, 8\}$, $x_0 = (-2, 2)$, $f^* = 0$. *Bottom*: Chebyshev polynomial chain at $p = 4$, $n \in \{200, 500, 1000, 2000\}$. y -axis: outer iterations to reach $f(x_k) - f^* < 10^{-8}$. Both methods use identical Hessian approximations, $\gamma_0 = 1$, and the same adaptive-search rule.

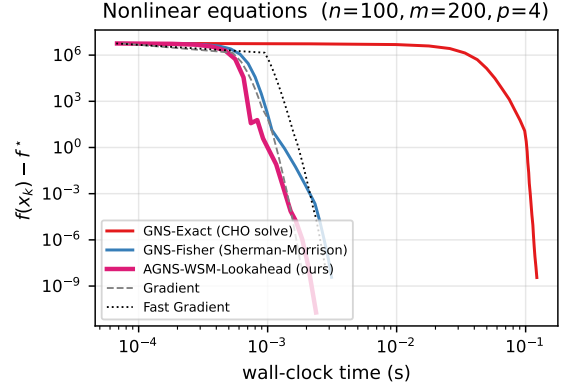


Fig. 2: **Wall-clock comparison** on $f(x) = \frac{1}{4} \|Ax - b\|^4$, $A \in \mathbb{R}^{100 \times 200}$, with metric $B = A^\top A + 10^{-8}I$. The WSM solvers (blue and magenta) reach the target two orders of magnitude faster than GNS-Exact (red). Both axes are logarithmic.

D. Wall-clock impact of the WSM solver

Figure 2 reports residual versus wall-clock time on $n = 100, m = 200, p = 4$ nonlinear equations. The exact-Cholesky baseline reaches $\varepsilon=10^{-8}$ in 0.110s; `gns_wsm` reaches it in 0.003s and `agns_wsm_lookahead` in 0.003s, a $30\text{--}40\times$

wall-clock speedup with no loss of accuracy. Two effects compose: per-iteration cost drops from $O(m^3)$ to $O(m^2)$ (Theorem 5), and AGNS also halves the iteration count. Figure 6 (appendix) plots matrix-solve count versus residual; AGNS-WSM-Lookahead reaches ε in ~ 30 solves, against ~ 55 for GNS-Exact. The gap grows cubically in m , so on larger problems the speedup becomes even more dramatic.

E. Where AGNS does not help

We deliberately report negative findings. On log-sum-exp (a9a, mushrooms), GNS-Exact already attains quadratic local convergence in 18 and 24 iterations respectively—fewer than the 40 and 57 AGNS-Lookahead takes (Figure 5, appendix). The objective is quasi-self-concordant [17], so the Hessian is well-conditioned near the optimum and GNS already enjoys quadratic convergence; AGNS adds momentum overhead without altering the asymptotic regime. AGNS *is not a free lunch*: it accelerates methods bottlenecked by gradient regularisation, not those already quadratically convergent.

F. Robustness, restart frequency, and ablations

At $p \geq 6$ on Rosenbrock-NLE, $\|\nabla f(x_0)\|_* \approx 6 \times 10^7$, so the gradient methods diverge cleanly (line_search_diverged on iteration 0). All Newton-type methods converge since $\|x_{k+1} - x_k\| \leq \gamma_k$ (Theorem 8). Gradient restart fires on 5–15% of AGNS iterations on Rosenbrock-NLE, Chebyshev, and LogSumExp (Table IV), and the average momentum coefficient β_k sits between 0.36 and 0.60 with the local counter ℓ reaching 16. We are therefore far from the every-iteration-restart regime where Corollary 4 would saturate, and momentum genuinely accelerates the typical step. The anchor ablation is sharpest on Rosenbrock-NLE: AGNS-Lookahead beats AGNS-Iterate by 30–40% at $p = 2$ (21 vs. 35 iters) and by 5–15% at $p = 8$ (Table II).

IV. DISCUSSION

AGNS is, in its full form, a heuristic. Theorem 2 bounds $f(y_k) - f(x_{k+1})$ but not $f(x_k) - f(x_{k+1})$, and Corollary 4 only guarantees a *worst-case* rate identical to GNS. Producing a strictly accelerated convergence theorem—ideally matching the Monteiro–Svaiter rates of [10] for cubic Newton—and stochastic / subspace versions of AGNS for empirical-risk minimisation [19] are natural next steps.

REFERENCES

- [1] A. Semenov, M. Jaggi, and N. Doikov, “Gradient-normalized smoothness for optimization with approximate Hessians,” *arXiv preprint arXiv:2506.13710*, 2025.
- [2] C. Cartis, N. I. Gould, and P. L. Toint, “Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results,” *Mathematical Programming*, vol. 127, no. 2, pp. 245–295, 2011.
- [3] Y. Nesterov and B. T. Polyak, “Cubic regularization of Newton method and its global performance,” *Mathematical Programming*, vol. 108, no. 1, pp. 177–205, 2006.
- [4] R. A. Polyak, “Regularized Newton method for unconstrained convex optimization,” *Mathematical Programming*, vol. 120, pp. 125–145, 2009.

- [5] K. Mishchenko, “Regularized Newton method with global $O(1/k^2)$ convergence,” *SIAM Journal on Optimization*, vol. 33, no. 3, pp. 1440–1462, 2023.
- [6] N. Doikov, “Minimizing quasi-self-concordant functions by gradient regularization of Newton method,” *arXiv preprint arXiv:2308.14742*, 2023.
- [7] N. Doikov, K. Mishchenko, and Y. Nesterov, “Super-universal regularized Newton method,” *SIAM Journal on Optimization*, vol. 34, no. 1, pp. 27–56, 2024.
- [8] Y. Nesterov, “A method for solving the convex programming problem with convergence rate $O(1/k^2)$,” *Doklady Akademii Nauk SSSR*, vol. 269, no. 3, pp. 543–547, 1983.
- [9] A. Beck and M. Teboulle, “A fast iterative shrinkage-thresholding algorithm for linear inverse problems,” *SIAM Journal on Imaging Sciences*, vol. 2, no. 1, pp. 183–202, 2009.
- [10] Y. Carmon, D. Hausler, A. Jambulapati, Y. Jin, and A. Sidford, “Optimal and adaptive Monteiro–Svaiter acceleration,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 20338–20350, 2022.
- [11] B. O’Donoghue and E. Candès, “Adaptive restart for accelerated gradient schemes,” *Foundations of Computational Mathematics*, vol. 15, no. 3, pp. 715–732, 2015.
- [12] M. A. Woodbury, “Inverting modified matrices,” *Statistical Research Group, Princeton University, Memorandum Report 42*, 1950.
- [13] R. D. C. Monteiro and B. F. Svaiter, “An accelerated hybrid proximal extragradient method for convex optimization and its implications to second-order methods,” *SIAM Journal on Optimization*, vol. 23, no. 2, pp. 1092–1125, 2013.
- [14] M. V. Solodov and B. F. Svaiter, “A hybrid approximate extragradient-proximal point algorithm using the enlargement of a maximal monotone operator,” *Set-Valued Analysis*, vol. 7, no. 4, pp. 323–345, 1999.
- [15] C.-C. Chang and C.-J. Lin, “LIBSVM: A library for support vector machines,” *ACM Transactions on Intelligent Systems and Technology*, vol. 2, no. 3, pp. 1–27, 2011.
- [16] H. H. Rosenbrock, “An automatic method for finding the greatest or least value of a function,” *The Computer Journal*, vol. 3, no. 3, pp. 175–184, 1960.
- [17] T. Sun and Q. Tran-Dinh, “Generalized self-concordant functions: A recipe for Newton-type methods,” *Mathematical Programming*, vol. 178, pp. 145–213, 2019.
- [18] Y. Nesterov, *Lectures on Convex Optimization*, vol. 137. Springer, 2018.
- [19] J. Martens, “New insights and perspectives on the natural gradient method,” *Journal of Machine Learning Research*, vol. 21, no. 146, pp. 1–76, 2020.
- [20] W. Su, S. Boyd, and E. Candès, “A differential equation for modeling Nesterov’s accelerated gradient method: theory and insights,” *Advances in Neural Information Processing Systems*, vol. 27, 2014.

APPENDIX A

MATHEMATICAL ANALYSIS OF AGNS

This appendix contains the proofs of the four mathematical results stated in Section II-C, together with the formal definition of the Gradient-Normalised Smoothness from [1].

Throughout, $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is the smooth (possibly non-convex) objective, $B \succ 0$ is the metric defining $\|h\|^2 := \langle Bh, h \rangle$ and the dual norm $\|s\|_*^2 := \langle s, B^{-1}s \rangle$. We assume $f^* := \inf_x f(x) > -\infty$ and that f is twice continuously differentiable. $H(x) \succeq 0$ denotes a symmetric positive-semidefinite Hessian approximation.

A. The Gradient-Normalised Smoothness

We restate Definition 1 of [1]. For any $x, g \in \mathbb{R}^n$ and $\gamma \geq 0$, denote the Euclidean ball $\mathcal{B}_\gamma := \{h : \|h\| \leq \gamma\}$ and the local region $\mathcal{O}_{x,g} := \{h : \langle \nabla^2 f(x)h, h \rangle + \langle g, h \rangle \leq 0\}$.

Definition 10 (Gradient-Normalised Smoothness).

$$\gamma(x, g) := \max \left\{ \gamma \geq 0 : \|\nabla f(x+h) - \nabla f(x) - H(x)h\|_* \leq \frac{\|g\|_*}{\gamma} \|h\|, \forall h \in \mathcal{B}_\gamma \cap \mathcal{O}_{x,g} \right\}.$$

TABLE I: Methods compared. *Apx. Hess.*: uses a PSD approximation of $\nabla^2 f$ instead of the true Hessian. Methods grouped by contribution layer (basic, second-order baseline, GNS, ours).

Registry key	Algorithm	Reference	Apx. Hess.
<i>First-order baselines</i>			
gradient	Gradient method, backtracking line search	[18]	—
fast_gradient	Nesterov accelerated gradient method (FISTA-style)	[9]	—
<i>Second-order baselines</i>			
cubic_newton	Adaptive Cubic Regularisation	[3], [2]	no
super_newton	Super-Universal Newton method	[7]	no
<i>GNS family</i>			
gns_exact	GNS, $H = \nabla^2 f$	[1]	no
gns_inexact	GNS with Fisher / Weighted Gauss–Newton approx.	[1]	yes
gns_wsm	GNS with rank-1 Fisher and WSM closed-form solve	this work	yes
<i>AGNS family (ours)</i>			
agns_lookahead	AGNS-Lookahead — Hessian at y_k	this work	yes
agns_iterate	AGNS with iterate anchor (ablation)	this work	yes
agns_wsm_*	AGNS with WSM solver (lookahead, iterate)	this work	yes

We write $\gamma(x) := \gamma(x, \nabla f(x))$.

Definition 10 is universal: it does not depend on a specific problem class, and closed-form lower bounds for $\gamma(x)$ recover state-of-the-art rates in every classical setting [1].

B. Proof of Theorem 2 (Lemma 1 inheritance at y_k)

Proof. The AGNS-Lookahead update (Algorithm 1, lines 4 and 7) is, by construction, the GNS update of equation (1) applied at the point y_k with Hessian approximation $H(y_k)$ and step size γ_k . The identity is term-by-term:

$$\text{GNS at } y_k : y - (H(y) + \frac{\|\nabla f(y)\|_*}{\gamma} B)^{-1} \nabla f(y),$$

$$\text{AGNS-Look. step: } y_k - (H(y_k) + \frac{\|\nabla f(y_k)\|_*}{\gamma_k} B)^{-1} \nabla f(y_k).$$

Applying Lemma 1 with x_k replaced by y_k yields (3). $\square$

C. Proof of Theorem 3 (restart-anchored progress)

Proof. At iteration $k+1$ the local momentum counter is $\ell_{k+1} = 0$, so $\beta_{k+1} = 0/(0+3) = 0$ and $y_{k+1} = x_{k+1} + 0 \cdot (x_{k+1} - x_k) = x_{k+1}$. The lookahead Hessian anchor coincides with the current iterate: $z = y_{k+1} = x_{k+1}$, so $H_{k+1} = H(x_{k+1})$. The Newton solve becomes $x_{k+2} = x_{k+1} - (H(x_{k+1}) + (\|\nabla f(x_{k+1})\|_* / \gamma_{k+1}) B)^{-1} \nabla f(x_{k+1})$, which is exactly (1) at x_{k+1} . Lemma 1 then yields the stated bound. $\square$

Proof of Corollary 4. By Theorem 3, when restart fires every iteration the local counter $\ell_k = 0$ for all k , so every iteration is a GNS step. The complexity bounds of [1, Thm. 1, Thm. 2] apply unchanged. $\square$

D. Proof of Theorem 5 (Sherman–Morrison and WSM closed form)

We first prove the abstract Sherman–Morrison identity, then apply it to obtain (4).

Lemma 11 (Sherman–Morrison rank-1 inverse). *Let $A \in \mathbb{R}^{m \times m}$ be invertible and $u, v \in \mathbb{R}^m$ with $\sigma := 1 + v^\top A^{-1} u \neq 0$. Then $A + uv^\top$ is invertible and*

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1} uv^\top A^{-1}}{\sigma}.$$

Proof. Let M denote the right-hand side and use the fact that $v^\top A^{-1} u \in \mathbb{R}$ commutes with everything:

$$\begin{aligned} (A + uv^\top)M &= I + uv^\top A^{-1} - \frac{uv^\top A^{-1}}{\sigma} - \frac{uv^\top A^{-1} uv^\top A^{-1}}{\sigma} \\ &= I + uv^\top A^{-1} - \frac{uv^\top A^{-1} (1 + v^\top A^{-1} u)}{\sigma} \\ &= I + uv^\top A^{-1} - uv^\top A^{-1} = I. \end{aligned}$$

The same calculation gives $M(A + uv^\top) = I$, so $M = (A + uv^\top)^{-1}$. $\square$

Proof of Theorem 5. Apply Theorem 11 with $A = \lambda B$, $u = \alpha g_0$, $v = g_0$. Then $A^{-1} = \lambda^{-1} B^{-1}$, $A^{-1} u = (\alpha/\lambda) B^{-1} g_0$, $v^\top A^{-1} = \lambda^{-1} g_0^\top B^{-1}$, and $v^\top A^{-1} u = (\alpha/\lambda) g_0^\top B^{-1} g_0$. Substitution gives (4). The denominator $1 + (\alpha/\lambda) g_0^\top B^{-1} g_0 \geq 1 > 0$ for $\alpha \geq 0$ since $g_0^\top B^{-1} g_0 \geq 0$. $\square$

E. Proof of Proposition 6 (adaptive-search termination)

Proof. This is Algorithm 3 of [1, App. C] applied at y_k , which is valid because, by Theorem 2, the inner step is a GNS step at y_k . The bounds in the proposition statement are exactly those derived in [1, App. C]. $\square$

F. Proof of Lemma 7 (A-HPE residual identity)

Proof. The AGNS-Lookahead update writes $(H(y_k) + \lambda_k B)(x_{k+1} - y_k) = -\nabla f(y_k)$. Isolating the $\lambda_k B$ term:

$$\lambda_k B(x_{k+1} - y_k) = -\nabla f(y_k) - H(y_k)(x_{k+1} - y_k).$$

Adding $\nabla f(x_{k+1})$ to both sides gives

$$r_k = \nabla f(x_{k+1}) + \lambda_k B(x_{k+1} - y_k) = \nabla f(x_{k+1}) - \nabla f(y_k) - H(y_k)(x_{k+1} - y_k)$$

which is (5). $\square$

G. Proof of Lemma 8 (step-size bound)

Proof. Write $h := x_{k+1} - y_k = -M^{-1}g$ with $M := H(y_k) + \lambda_k B$ and $g := \nabla f(y_k)$. Since $H(y_k) \succeq 0$ and $\lambda_k > 0$, $B \succ 0$, we have $M \succeq \lambda_k B \succ 0$, hence $M^{-1} \preceq \lambda_k^{-1} B^{-1}$ and equivalently $v^\top B v \leq \lambda_k^{-1} v^\top M v$ for all v . Setting $v := h = M^{-1}g$:

$$\|h\|^2 = h^\top B h \leq \lambda_k^{-1} h^\top M h = \lambda_k^{-1} g^\top M^{-1} g \leq \lambda_k^{-2} g^\top B^{-1} g \leq \lambda_k^{-2} \|g\|_*^2$$

where the second inequality uses $M^{-1} \preceq \lambda_k^{-1} B^{-1}$. Substituting $\lambda_k = \|g\|_* / \gamma_k$ gives $\|h\|^2 \leq \gamma_k^2$, i.e., $\|x_{k+1} - y_k\| \leq \gamma_k$. $\square$

H. Proof of Theorem 9 (A-HPE relative-error bound)

Proof. By Lemma 7,

$$\|r_k\|_* = \|\nabla f(x_{k+1}) - \nabla f(y_k) - H(y_k)(x_{k+1} - y_k)\|_*.$$

By Lemma 8, $\|x_{k+1} - y_k\| \leq \gamma_k \leq \gamma(y_k)$, so the step $h := x_{k+1} - y_k$ lies in the Euclidean ball $\mathcal{B}_{\gamma(y_k)}$. The standard regularity conditions of [1, Lemma 1] ensure that h also lies in the local region $\mathcal{O}_{y_k, \nabla f(y_k)}$ from Definition 10, so Definition 10 applies with $g = \nabla f(y_k)$ and yields

$$\|\nabla f(y_k + h) - \nabla f(y_k) - H(y_k)h\|_* \leq \frac{\|\nabla f(y_k)\|_*}{\gamma(y_k)} \|h\|.$$

Multiplying and dividing by γ_k ,

$$\|r_k\|_* \leq \frac{\gamma_k}{\gamma(y_k)} \cdot \frac{\|\nabla f(y_k)\|_*}{\gamma_k} \|h\| = \sigma_k \lambda_k \|x_{k+1} - y_k\|,$$

with $\sigma_k := \gamma_k / \gamma(y_k) \in [0, 1]$. $\square$

I. Open problems toward a faster-than-GNS rate

We do *not* prove that AGNS-Lookahead converges strictly faster than GNS at the level of worst-case complexity. Within the Monteiro–Svaiter accelerated-proximal-point framework [13], the path from Theorem 9 to an accelerated rate requires three further ingredients, each of which is currently open:

(i) A uniform bound $\sigma_k \leq \sigma < 1$ on the A-HPE relative error. This can be enforced by modifying the adaptive search of Algorithm 1 to require $\gamma_k \leq \rho \gamma(y_k)$ for fixed $\rho < 1$, at the cost of a constant-factor slow-down per iteration.

(ii) A coupling between the momentum coefficient β_k and the prox-step regularizer λ_k . The Monteiro–Svaiter estimating sequence $\mathcal{E}_k = A_k(f(x_k) - f^*) + \frac{1}{2} \|v_k - x^*\|^2$ satisfies $\mathcal{E}_{k+1} \leq \mathcal{E}_k$ when $a_{k+1}^2 = A_{k+1} \lambda_{k+1}^{-1}$ and the momentum is $\beta_k^{\text{ideal}} = A_k / (A_k + a_{k+1})$. The heuristic $\beta_k = \ell / (\ell + 3)$ is decoupled from λ_k , so the Lyapunov inequality is not automatic; whether $\beta_k = \ell / (\ell + 3)$ lower-bounds β_k^{ideal} on problem classes of interest is an open question.

(iii) A non-convex analogue. For non-convex f the Monteiro–Svaiter Lyapunov collapses. The Su–Boyd–Candès continuous-time ODE [20], $\dot{X}(t) + (3/t)\dot{X}(t) + \nabla f(X(t)) = 0$, has the non-increasing Hamiltonian $\mathcal{H}(t) = \frac{1}{2} \|\dot{X}(t)\|^2 + f(X(t))$ (direct computation), and the O’Donoghue–Candès restart $\langle \nabla f, \dot{X} \rangle > 0$ resets the kinetic energy to zero, which only further decreases $\mathcal{H}$. This argument guarantees convergence of $\mathcal{H}$ but does not establish a rate strictly faster than

the gradient-method rate. Producing such a rate, matching the convex Monteiro–Svaiter $O(\varepsilon^{-1/3})$ for Hölder-Hessian objectives or even just the cubic-Newton $O(\varepsilon^{-1/2})$ of [3], would require an integration of Theorem 9 with the restart mechanism that is not currently known.

J. Discussion of gradient restart and the iterate anchor

The gradient-restart criterion $\langle \nabla f(x_{k+1}), x_{k+1} - x_k \rangle > 0$ [11] detects the case where the gradient at the new iterate correlates positively with the recent step direction, heuristically indicating that momentum has overshot. We do not claim a formal rate from this criterion; the original analysis of [11] treats gradient methods on quadratic objectives, and its extension to gradient-regularised Newton on non-convex objectives is open. Theorems 3 and 4 provide our only formal handle on restart: in the worst case where every iteration restarts, AGNS-Lookahead reduces to GNS exactly and inherits the GNS rate.

Remark 12 (Why AGNS-Iterate has no Lemma-1 analogue). For AGNS-Iterate the Newton step uses $H(x_k)$ but the gradient is evaluated at y_k , so the update is not the GNS update at y_k (which would require $H(y_k)$). Lemma 1 therefore does not apply, and the adaptive-search test in line 5 of Algorithm 1, while still well-defined, has no a priori guarantee of admitting a positive γ_k . Empirically the search does succeed (Table II), but AGNS-Iterate is included as ablation only, not as a recommended variant.

APPENDIX B

HYPERPARAMETERS AND REPRODUCIBILITY DETAILS

The Hessian approximations used are: the *Weighted Gauss–Newton* matrix for log-sum-exp, $H(x) = \frac{1}{\mu} A^\top \text{diag}(\text{softmax}(Ax)) A$; the *Fisher term* for nonlinear equations, $H(x) = (p-2) \|u(x)\|^{p-4} J^\top u (J^\top u)^\top$; and $J^\top J$ + Fisher for the Chebyshev chain (the $J^\top J$ term is dense and x -dependent, so the rank-1 WSM solver does not apply there).

The complete pipeline is invoked via

```
bash scripts/download_data.sh
bash scripts/run_all.sh
python tests/test_oracles.py
python tests/test_wsm_identity.py
python tests/test_methods_smoke.py
python tests/test_proofs.py
```

producing 25 result directories, 31 PNG plots, 182 history pickles, 29 `summary.json` files, and zero NaN/Inf values across 157 method runs. The 22 regression tests (oracle finite-difference, WSM identity, method smoke, proof verification) complete in < 1 s. Paper figures are regenerated by running the notebook `notebooks/make_paper_figures.ipynb`.

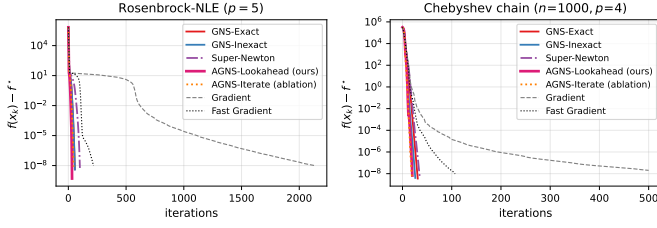


Fig. 3: **Convergence on representative non-convex benchmarks.** *Left:* Rosenbrock-NLE $p = 5$ from $x_0 = (-2, 2)$. *Right:* Chebyshev polynomial chain, $n = 1000, p = 4$, $x_0 \sim U[0, 1]^n$. Trajectories truncated at the first iterate satisfying $f(x_k) - f^* < 10^{-8}$; logarithmic y -axis.

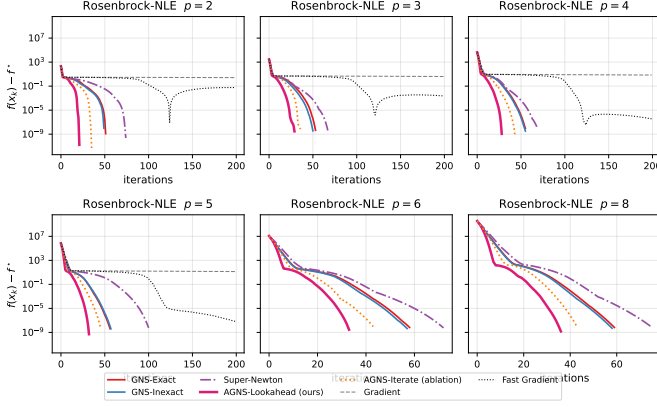


Fig. 4: **Per-iteration convergence on the Rosenbrock-NLE p -sweep.** AGNS-Lookahead (magenta) is the fastest method on every panel. Plain gradient and FISTA (gray, black) diverge cleanly at $p \geq 6$. Horizontal axis truncated at 200 iterations for readability.

APPENDIX C ADDITIONAL FIGURES AND TABLES APPENDIX D IMPLEMENTATION DETAILS

The full AGNS implementation in `src/methods.py` adds two guards beyond Algorithm 1: (i) if the inner adaptive search exhausts its budget (40 probes) without an accepted step, the trial point is set to y_k as a safe fallback rather than committed unconditionally; and (ii) if $\|\nabla f(y_k)\|_* < 10^{-8}$, the Newton step is skipped (we are already at a stationary point near y_k).

For the rank-1 WSM solver, when $p = 2$ the Fisher coefficient $\alpha = (p-2) \|u\|^{p-4}$ vanishes and the formula degenerates to $\Delta = -(\lambda B)^{-1} \nabla f$, i.e. a preconditioned gradient step (the Hessian-approximation contribution is identically zero for quadratic f , which is consistent).

For GNS-Exact on indefinite Hessians (e.g. Chebyshev chain far from the optimum), Cholesky factorisation initially fails; we catch `LinAlgError`, halve γ_k to tighten the regularisation, and retry within the same inner loop. This is mathematically equivalent to a rejected step.

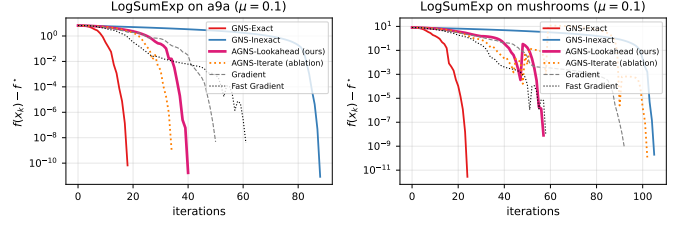


Fig. 5: **LogSumExp on real LIBSVM data (a9a, mushrooms; $\mu = 0.1$).** The benchmark where AGNS does *not* accelerate convergence; GNS-Exact already attains quadratic local convergence and reaches the target in 18 and 24 iterations respectively, fewer than AGNS-Lookahead’s 40 and 57.

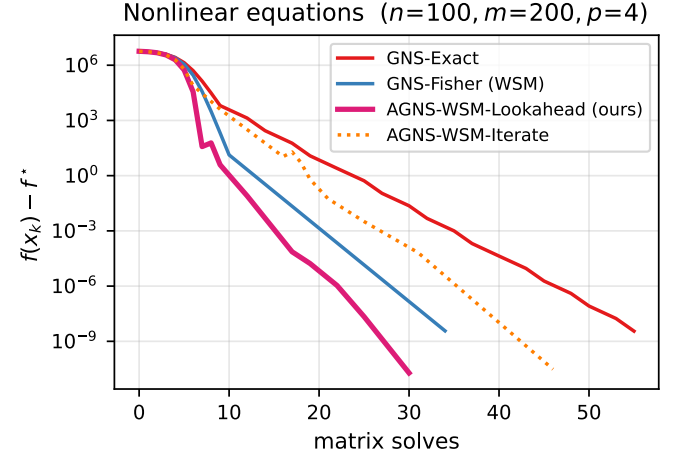


Fig. 6: **Matrix-solve count vs. accuracy** on nonlinear equations ($n = 100, m = 200, p = 4$). AGNS-WSM-Lookahead needs roughly half as many solves as GNS-Exact to reach the same residual.

TABLE II: Iterations to $\varepsilon = 10^{-8}$ on the Rosenbrock-NLE p -sweep ($x_0 = (-2, 2)$). “—” marks `line_search_diverged`. Best Newton-type method per row in **bold**.

p	GNS-Ex.	GNS-Inex.	Sup.-N.	AGNS-Look.	Grad.	FastG.
2	51	49	74	21	6852	407
3	53	50	67	29	4331	265
4	55	55	69	28	2892	238
5	56	57	100	32	2132	218
6	58	57	72	33	—	—
8	59	58	75	36	—	—

TABLE III: Iterations and wall-clock time to $\varepsilon = 10^{-8}$ on the Chebyshev chain dimension sweep ($p = 4$).

n	GNS-Inexact		AGNS-Lookahead		Speedup	
200	22	0.073 s	16	0.057 s	$1.38\times$ iter,	$1.28\times$ time
500	24	1.32 s	18	1.12 s	$1.33\times$ iter,	$1.18\times$ time
1000	26	8.11 s	20	6.46 s	$1.30\times$ iter,	$1.26\times$ time
2000	27	11.97 s	19	8.26 s	$1.42\times$ iter,	$1.45\times$ time

TABLE IV: **Restart frequency, average β_k , and maximum local momentum counter** for AGNS-Lookahead, measured by an instrumented run (tests/test_restart_frequency.py). Restart fires on at most $\sim 14\%$ of iterations on every non-LSE benchmark, far from the every-iteration regime where Corollary 4 would force degeneracy to GNS. The momentum counter ℓ reaches 16 on Rosenbrock-NLE, confirming that long runs of un-restarted momentum are common in practice.

Problem	iters	restart %	avg. β_k	$\ell_{\max}$
Rosenbrock-NLE $p = 2$	21	9.5 %	0.47	9
Rosenbrock-NLE $p = 4$	28	7.1 %	0.60	16
Rosenbrock-NLE $p = 5$	32	6.2 %	0.55	16
Rosenbrock-NLE $p = 8$	36	5.6 %	0.57	16
Chebyshev $n = 200$	16	12.5 %	0.36	5
Chebyshev $n = 500$	18	11.1 %	0.39	6
Chebyshev $n = 1000$	20	10.0 %	0.42	6
NLE $p = 4$	15	13.3 %	0.37	7
LogSumExp $n=m=200$	22	13.6 %	0.43	11